

DAY 04

Intro to the CLI

28 source flashcards | Cisco IOS modes, commands, configs & passwords



```
Router> enable
```

```
Router# configure terminal
```

```
Router(config)# hostname R1
```

```
R1(config)# exit
```

```
R1# copy running-config startup-config
```

QUESTION

What two kinds of console ports are common on Cisco devices?

COMMON CONSOLE PORTS**USB**

modern console connection

RJ45

traditional console port

ANSWER

USB
RJ45

Cisco devices commonly provide both a USB console port and an RJ45 console port for local management.

QUESTION

What kind of cable is used to connect to the RJ45 console port on a Cisco device?

RJ45 CONSOLE CONNECTION

ROLLOVER CABLE

PC / terminal

Cisco console

Pin order is reversed end-to-end

1 2 3 4 5 6 7 8 -> 8 7 6 5 4 3 2 1

ANSWER

Rollover cable

A rollover cable connects a computer/terminal to the RJ45 console port. Its pin order is reversed end-to-end.

QUESTION

Default Cisco console port settings: Speed (baud rate) = [...]

Cisco console defaults

Speed	9600 bps
-------	----------

Data bits	8
-----------	---

Parity	None
--------	------

Stop bits	1
-----------	---

Flow control	None
--------------	------

Memory cue: 9600 - 8 - N - 1, flow control None

ANSWER

9600 bits per second

The default console line speed is 9600 bits per second (baud).

QUESTION

Default Cisco console port settings: Data bits = [...]

Cisco console defaults

Speed 9600 bps

Data bits 8

Parity None

Stop bits 1

Flow control None

Memory cue: 9600 - 8 - N - 1, flow control None

ANSWER

8

Cisco console sessions use 8 data bits by default.

QUESTION

Default Cisco console port settings: Stop bits = [...]

Cisco console defaults

Speed 9600 bps

Data bits 8

Parity None

Stop bits 1

Flow control None

Memory cue: 9600 - 8 - N - 1, flow control None

ANSWER

1

Cisco console sessions use 1 stop bit by default.

QUESTION

Default Cisco console port settings: Parity = [...]

Cisco console defaults

Speed 9600 bps

Data bits 8

Parity None

Stop bits 1

Flow control None

Memory cue: 9600 - 8 - N - 1, flow control None

ANSWER

None

The default parity setting is None.

QUESTION

Default Cisco console port settings: Flow control = [...]

Cisco console defaults

Speed	9600 bps
Data bits	8
Parity	None
Stop bits	1

Flow control

Memory cue: 9600 - 8 - N - 1, flow control None

None

ANSWER

None

The default flow-control setting is None.

QUESTION

In Cisco IOS, user EXEC mode is indicated by what?
Router[...]

USER EXEC MODE



ANSWER

>

The > prompt means user EXEC mode, the lowest IOS access level.

QUESTION

In Cisco IOS, privileged EXEC mode is indicated by what?
Router[...]

PRIVILEGED EXEC MODE



```
Router#
```

ANSWER

#

The # prompt means privileged EXEC mode, which provides much broader access.

QUESTION

In Cisco IOS, global configuration mode is indicated by what?
Router[...]

GLOBAL CONFIGURATION MODE



```
Router(config)#
```

ANSWER

(config)#

The (config)# prompt means global configuration mode.

QUESTION

What is the lowest-level mode in Cisco IOS?

IOS PRIVILEGE LADDER

Global config

Router(config)#

Privileged EXEC

Router#

User EXEC

Router>

ANSWER

user EXEC mode

User EXEC is the starting and lowest-level IOS mode.

QUESTION

In Cisco IOS, what mode provides complete access to view the device's configuration files?

IOS PRIVILEGE LADDER

Global config

Router(config)#

Privileged EXEC

Router#

User EXEC

Router>

ANSWER

privileged EXEC mode

Privileged EXEC mode lets you view the complete configuration and enter configuration modes.

QUESTION

View the available commands:
Router#[...]



```
Router# ?
```

```
show      Show running system information  
configure Enter configuration mode
```

ANSWER

?

The ? key displays context-sensitive help and available commands.

QUESTION

Enter privileged EXEC mode:
Router>[...]



```
Router> enable
```

```
Router#
```

ANSWER

enable

Use enable from user EXEC mode to enter privileged EXEC mode.

QUESTION

Enter global configuration mode:
Router# [...]



```
Router# configure terminal  
Router(config)#
```

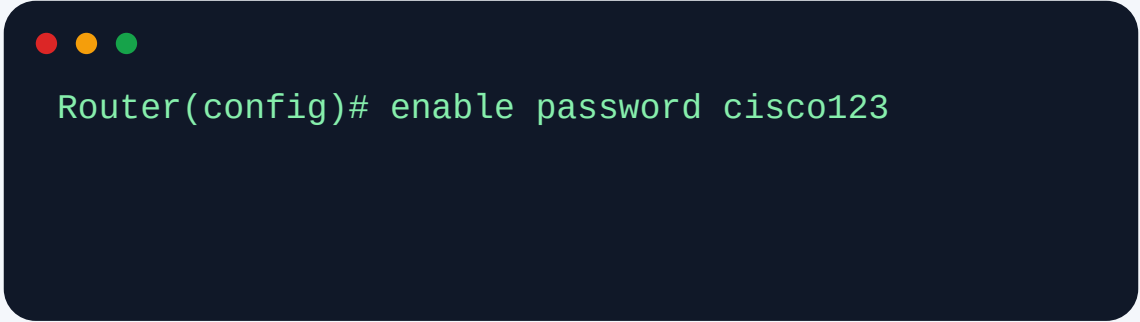
ANSWER

configure terminal

Use configure terminal from privileged EXEC mode to enter global configuration mode.

QUESTION

Configure an unencrypted password to protect privileged EXEC mode:
Router(config)# [...]



```
Router(config)# enable password cisco123
```

UNENCRYPTED unless password encryption is enabled

ANSWER

enable password password

enable password sets an unencrypted privileged EXEC password unless password encryption is separately enabled.

QUESTION

Return to privileged EXEC mode:
Router(config)# [...]



```
Router(config)# exit
```

```
Router#
```

ANSWER

exit

exit moves back one IOS configuration level; here it returns to privileged EXEC mode.

QUESTION

What two configuration files are kept on the device?

TWO CONFIGURATION FILES**running-config**

active now
stored in RAM

startup-config

loaded at boot
stored in NVRAM

ANSWER

running-config, startup-config

running-config is the active configuration in RAM; startup-config is the saved configuration used at boot.

QUESTION

**View the running configuration:
Router# [...]**



```
Router# show running-config  
... configuration output ...
```

ANSWER

show running-config

show running-config displays the active configuration currently in use.

QUESTION

**View the startup configuration:
Router# [...]**



```
Router# show startup-config  
... configuration output ...
```

ANSWER

show startup-config

show startup-config displays the saved configuration that will be loaded at startup.

QUESTION

Which three commands can be use to save the configuration?

Router# [...]

Router# [...]

Router# [...]



```
Router# write
```

```
Router# write memory
```

```
Router# copy running-config startup-config
```

All three save the configuration

ANSWER

write

write memory

copy running-config startup-config

All three commands save the running configuration so it survives a reload.

QUESTION

**Encrypt current and future passwords on the device:
Router(config)# [...]**



```
Router(config)# service password-encryption
```

plain text -> encrypted display

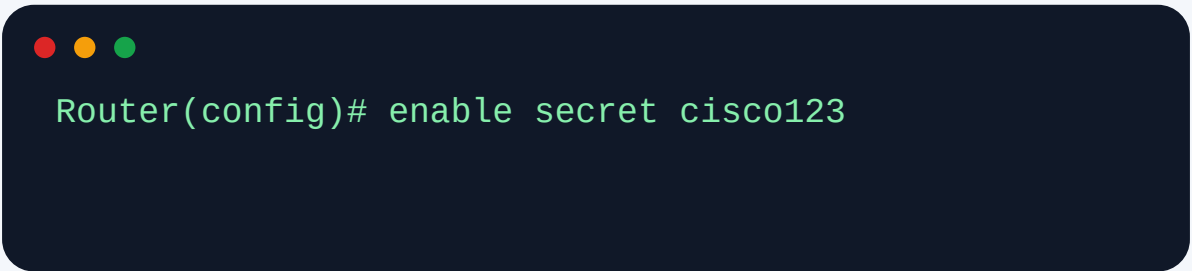
ANSWER

service password-encryption

service password-encryption encrypts current and future plain-text passwords stored in the configuration.

QUESTION

**Configure an MD5-encrypted enable password:
Router(config)# [...]**



```
Router(config)# enable secret cisco123
```

Protected enable secret

ANSWER

enable secret password

enable secret creates the stronger protected privileged EXEC password and historically uses type 5 MD5 in this course material.

QUESTION

You use the [...] keyword to execute privileged EXEC commands in global configuration mode.



```
Router(config)# do show running-config  
... output ...
```

Run EXEC commands without leaving config mode

ANSWER

do

The do keyword lets you run privileged EXEC commands without leaving global configuration mode.

QUESTION

Use the [...] keyword to remove a configured command.



```
Router(config)# service password-encryption  
Router(config)# no service password-encryption
```

no = remove / negate the configuration command

ANSWER

no (for example, no service password-encryption)

Prefix a configuration command with no to remove or negate that configuration.

QUESTION

Cisco's service password-encryption command uses type [...] encryption.

CISCO PASSWORD TYPE

TYPE 7

service password-encryption

ANSWER

7

service password-encryption uses Cisco type 7 reversible obfuscation, not strong cryptographic protection.

QUESTION

enable secret uses type [...] (MD5) encryption.

CISCO PASSWORD TYPE

TYPE 5**enable secret (MD5 in this material)**

ANSWER

5

In this course material, enable secret is represented as type 5 MD5 encryption.

QUESTION

If service password-encryption is removed, will currently encrypted passwords be decrypted?

ENCRYPTED STAYS ENCRYPTED

Removing the command does not decrypt existing entries

ANSWER

No

Removing service password-encryption does not decrypt passwords that are already stored in encrypted form.

DAY 04 - QUICK REVIEW

Console defaults

9600 - 8 - N - 1

Speed | Data bits | Parity | Stop bits + Flow control: None

IOS mode ladder

User EXEC Router> enable

Privileged EXEC Router# configure terminal

Global configuration Router (config)# exit

Key commands

Help ?	Encrypt passwords service password-encryption
Show active config show running-config	Protected enable password enable secret password
Show saved config show startup-config	Run EXEC in config mode do ...
Save copy running-config startup-config	Remove config no ...